

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE-AI	Date:	17-08-2026

Code of Conduct

/ G.Vinay Kumar Reddy(192472093) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as “best hotels near...” and “book a flight from Chennai to...”. The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as “give the book to John,” “sleep,” and “put the file on the desk.” The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

QUESTION 1 – CONSTITUENCY PARSING VS DEPENDENCY PARSING

Scenario

A conversational AI system must identify grammatical relationships in queries containing long-distance dependencies. The developers compare constituency parsing and dependency parsing.

Constituency Parsing

Constituency parsing represents a sentence as a hierarchy of phrases. Words are grouped into units such as noun phrases (NP), verb phrases (VP), and prepositional phrases (PP). This makes the internal structure of a sentence easy to examine.

Dependency Parsing

Dependency parsing represents a sentence through direct relationships between words. A word acts as a head and other words depend on it through relations such as subject, object, and modifier.

Comparison

Constituency parsing is better for showing phrase boundaries and nested structure. Dependency parsing is more direct when the system needs grammatical relationships between individual words.

Example

In “The student who submitted the assignment thanked the teacher,” dependency parsing can identify “thanked” as the main verb, “student” as the subject, and “teacher” as the object.

Critical Analysis

For conversational AI, direct word relationships are often easier to use for information extraction and semantic processing. Constituency structure is still valuable when phrase-level structure is required.

Final Choice

Dependency parsing is preferred for the stated scenario because it gives a compact representation of grammatical relationships and is practical for downstream conversational tasks.

QUESTION 2 – TOP-DOWN PARSING VS EARLEY PARSING

Scenario

The system processes incomplete queries such as “best hotels near...” and “book a flight from Chennai to...”. It currently uses top-down parsing and is considering Earley parsing.

Top-Down Parsing

Top-down parsing begins with the start symbol and predicts grammar rules to derive the input. It is simple and works well when the grammar and expected sentence structure are predictable.

Earley Parsing

Earley parsing is a chart-parsing method for general context-free grammars. It maintains partial parsing states through prediction, scanning, and completion.

Comparison

Top-down parsing can become less convenient when the input is incomplete because the final structure is not known. Earley parsing can preserve partial analyses and continue them when more words arrive.

Example

For “book a flight from Chennai to...”, Earley parsing can keep a partial state expecting a destination. After the destination is entered, the parser can complete the existing analysis.

Critical Analysis

Interactive search and voice systems receive input incrementally and may also contain ambiguity. Earley parsing is therefore more flexible, although a restricted top-down parser can be simpler for a small fixed grammar.

Final Choice

Earley parsing is more suitable for this scenario because it handles incomplete and ambiguous input while maintaining partial parsing information.

QUESTION 3 – CFG VS PCFG VS NEURAL CONSTITUENCY PARSING

Scenario

A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. CFG, PCFG, and neural constituency parsing are compared.

CFG

A Context-Free Grammar represents possible syntactic structures through rules such as $S \rightarrow NP VP$. It is clear and rule-based, but a basic CFG does not assign probabilities to competing interpretations.

PCFG

A Probabilistic Context-Free Grammar adds probabilities to grammar rules. These probabilities allow alternative parses to be ranked and the more probable interpretation to be selected.

Neural Constituency Parsing

A neural constituency parser learns contextual patterns from training data and predicts phrase structures. It can capture information that is difficult to encode with manually written rules.

Comparison

CFG has strong interpretability but does not naturally resolve ambiguity. PCFG preserves explicit grammar rules while adding probabilistic ranking. Neural parsing can provide strong practical performance but is less transparent.

Evidence / Example

In “I saw the man with a telescope,” the phrase “with a telescope” can have different attachments. A PCFG can use learned rule probabilities to rank the possible structures.

Critical Analysis

If explainability and a principled method for choosing among parses are important, PCFG provides a useful balance. If maximum data-driven accuracy is the priority and annotated data are available, a neural parser may be preferable.

Final Choice

For the stated grammar-checking requirement, PCFG is a strong choice because it combines understandable grammar rules with probabilistic ambiguity resolution.

QUESTION 4 – FEATURE STRUCTURES VS TRADITIONAL CFG RULES

Scenario

A multilingual NLP system must process number, gender, tense, and person. Feature structures are compared with traditional context-free grammar rules.

Traditional CFG

A traditional CFG describes basic sentence structure using rules such as $S \rightarrow NP VP$. Basic CFG rules do not directly represent agreement information such as singular, plural, person, or gender.

Feature Structures

Feature structures attach attribute-value information to linguistic units. Typical attributes include NUMBER, GENDER, PERSON, TENSE, and CASE.

Feature Unification

Feature unification checks whether feature values are compatible. For example, a singular subject can be required to combine with a compatible singular verb. Conflicting values can cause the analysis to fail.

Comparison

CFG provides the structural skeleton of a sentence. Feature structures add grammatical constraints to that structure and can avoid creating many separate rules for every feature combination.

Multilingual Evidence

Different languages may express gender, number, person, tense, or case differently. Feature structures provide a flexible way to represent these properties while retaining a common grammatical framework.

Critical Analysis

Feature structures require more representation and processing than a simple CFG, but the additional information is valuable when grammatical agreement is important.

Final Choice

Feature structures are more suitable for the multilingual scenario because they explicitly represent and enforce the required grammatical features.

QUESTION 5 – SUBCATEGORIZATION FRAMES

Scenario

A voice assistant must interpret commands such as “give the book to John,” “sleep,” and “put the file on the desk.” The verbs have different argument requirements.

Meaning

Subcategorization describes the arguments or complements that a verb requires or permits. This lexical information helps the parser predict what should occur with a verb.

Example – Give

The verb “give” generally involves a giver, an item, and a recipient. In “Give the book to John,” “book” is the item and “John” is the recipient.

Example – Sleep

The verb “sleep” is generally intransitive and does not require a direct object. “The child sleeps” is therefore a complete structure.

Example – Put

The verb “put” normally requires an object and a location. In “Put the file on the desk,” “file” is the object and “on the desk” supplies the location.

Comparison / Analysis

Treating all verbs as having the same structure would cause incorrect interpretations. Subcategorization distinguishes verbs according to their required arguments and complements.

Voice Assistant Application

If the user says “Put the file,” the system can detect that the required location is missing and ask a focused clarification question such as “Where should I put the file?”

Critical Analysis

Subcategorization supports both syntactic and semantic interpretation because the arguments can be associated with roles such as theme, recipient, and location.

Final Choice

Subcategorization frames are important for the voice assistant because they improve command interpretation and help identify missing or unsuitable arguments.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1	3	3	3	3	3	15	75
2	3	3	3	3	3	15	
3	3	3	3	3	3	15	
4	3	3	3	3	3	15	
5	3	3	3	3	3	15	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____